

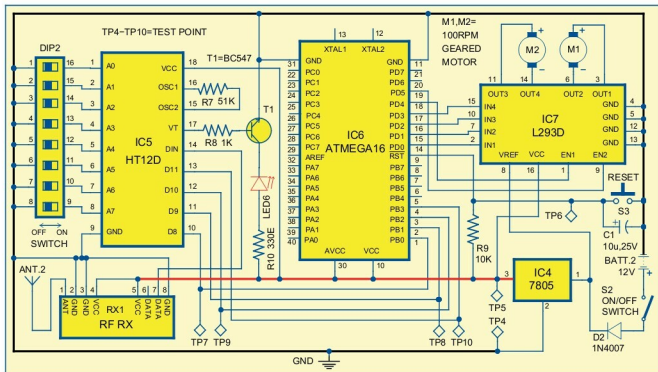


Fig. 4: Circuit for the robot

crosses the pre-defined voltage level, the corresponding comparator's output goes low. The same is also indicated by LED1 through LED4. The outputs of the comparators are fed to inputs AD8 through AD11 of IC1. The generated

PARTS LIST

Semiconductors:

IC1	- HT12E encoder
IC2	- LM339 comparator
IC3, IC4	- 7805, 5V regulator
IC5	- HT12D decoder
IC6	- ATmega16 microcontroller
IC7	- L293D motor driver
T1	- BC547 npn transistor
LED1-LED6	- 5mm LED
D1, D2	- 1N4007 rectifier diode
TX1	- ASK transmitter module
RX1	- ASK receiver module
	- Accelerometer module

Resistors (all 1/4-watt, ±5 per cent carbon):

R1	- 1-mega-ohm
R2-R5	- 470-ohm
R6, R10	- 330-ohm
R7	- 51-kilo-ohm
R8	- 1-kilo-ohm
R9	- 10-kilo-ohm
VR1-VR4	- 10-kilo-ohm preset

Capacitors:

C1	- 10µF, 25V electrolytic
----	--------------------------

Miscellaneous:

DIP1, DIP2	- 16-pin DIP switch
S1, S2	- On/off switch
S3	- Push-to-on switch
M1, M2	- 12V, 100-rpm geared motor
BATT.1	- 9V battery (PP3)
BATT.2	- 12V, 4.5Ah battery
ANT.1-ANT.2	- 10 cm hook-up wire

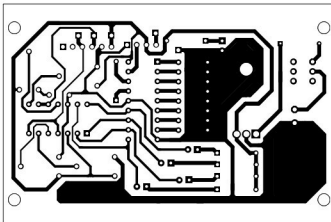


Fig. 5: An actual-size, single-side PCB for the wireless controller

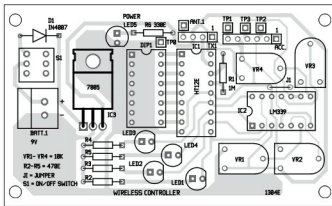


Fig. 6: Component layout for the PCB in Fig. 5

code is encoded through encoder IC1 and transmitted using the RF transmitter module TX1. Regulator IC3 provides regulated 5V supply to the circuit.

Accelerometer module. An accelerometer is an electromechanical device that measures acceleration of anything that it is mounted on. The accelerometer module used here is based on ADXL335 triple-axis accelerometer from Analog Devices. The sensor has a full sensing range of $\pm 3g$.

Robot. It comprises an